E1. Erase and Extend (Easy Version)

time limit per test: 2 seconds

memory limit per test: 256 megabytes

This is the easy version of the problem. The only difference is the constraints on n and k. You can make hacks only if all versions of the problem are solved.

You have a string s, and you can do two types of operations on it:

- Delete the last character of the string.
- Duplicate the string: $s := s + s$, where $+$ denotes concatenation.

You can use each operation any number of times (possibly none).

Your task is to find the lexicographically smallest string of length exactly k that can be obtained by doing these operations on string s.

A string a is lexicographically smaller than a string b if and only if one of the following holds:

- a is a prefix of b, but $a \neq b$;
- In the first position where a and b differ, the string a has a letter that appears earlier in the alphabet than the corresponding letter in b.

Input

The first line contains two integers n, k ($1 \leq n, k \leq 5000$) — the length of the original string s and the length of the desired string.

The second line contains the string s, consisting of n lowercase English letters.

Output

Print the lexicographically smallest string of length k that can be obtained by doing the operations on string s.

Examples

input

```
8 16
dbcadabc
```

output

```
dbcadabcdbcadabc
```

input

```
4 5
abcd
```

output

```
aaaaa
```

Note

In the first test, it is optimal to make one duplication: "dbcadabc" $\to$ "dbcadabcdbcadabc".

In the second test it is optimal to delete the last 3 characters, then duplicate the string 3 times, then delete the last 3 characters to make the string have length k.

"abcd" $\to$ "abc" $\to$ "ab" $\to$ "a" $\to$ "aa" $\to$ "aaaa" $\to$ "aaaaaaaa" $\to$ "aaaaaaa" $\to$ "aaaaaa" $\to$ "aaaaa".